**Turing Test –CAPTCHA – Suggested Architecture**

**CAPTCHA = Completely Automated Public Turing test to tell Computers and Humans Apart.**

**High-Level Architecture**

**Components:**

1. **Challenge Generator**
    - **Text distortion**
    - **Image selection**
    - **Puzzle generation**
2. **Frontend Interface**
    - **Displays CAPTCHA**
    - **Accepts user input**
3. **Verification Engine**
    - **Compares input with stored answer**
    - **Time threshold check**
4. **Security Module**
    - **Rate limiting**
    - **Bot detection heuristics**
    - **IP logging**
5. **Database**
    - **Stores challenges**
    - **Stores hash of answers**

**CAPTCHA – Architecture**

CAPTCHA = Completely Automated Public Turing test to tell Computers and Humans Apart.

**High-Level Architecture**

**Components:**

1. **Challenge Generator**
    - Text distortion
    - Image selection
    - Puzzle generation
2. **Frontend Interface**
    - Displays CAPTCHA

    - Accepts user input
3. **Verification Engine**
    - Compares input with stored answer
    - Time threshold check
4. **Security Module**
    - Rate limiting
    - Bot detection heuristics
    - IP logging
5. **Database**
    - Stores challenges
    - Stores hash of answers

**Difference Between BFS and DFS**

| **Feature** | **BFS** | **DFS** |
|---|---|---|
| Data Structure | Queue | Stack |
| Complete | Yes | Not always |
| Optimal | Yes | No |
| Space Required | High | Low |
| Time Complexity | O(b^d) | O(b^m) |

Where:

- b = branching factor
- d = depth of solution
- m = max depth